

# Lecture 2: Density Functional Theory

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## 1. The Idea

Hartree–Fock gave us a mean-field theory with no correlation. Density functional theory keeps the mean-field cost but promises, in principle, the *exact* ground-state energy. Three things are worth saying up front: DFT is the most widely used electronic structure method by an enormous margin; it is exact in principle but not exact in practice even for one electron; and it comes in hundreds of flavors, which is a symptom of the fact that nobody knows the exact functional.

The motivation is dimensional. The wavefunction needs  $\mathcal{O}(N_g^{3N})$  numbers. But the Hamiltonian contains only one- and two-body operators, so the energy needs far less. Define the one- and two-particle reduced density matrices

$$\gamma(\mathbf{r}'_1, \mathbf{r}_1) = N \int d\mathbf{r}_2 \cdots d\mathbf{r}_N \Psi^*(\mathbf{r}'_1, \mathbf{r}_2, \dots) \Psi(\mathbf{r}_1, \mathbf{r}_2, \dots), \quad (1)$$

$$\Gamma(\mathbf{r}_1, \mathbf{r}_2) = \frac{N(N-1)}{2} \int d\mathbf{r}_3 \cdots d\mathbf{r}_N |\Psi(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3, \dots)|^2, \quad (2)$$

with the electron density as the diagonal of the 1-RDM,  $\rho(\mathbf{r}) = \gamma(\mathbf{r}, \mathbf{r})$  and  $\int \rho d\mathbf{r} = N$ . Then *exactly*

$$E = \int d\mathbf{r} d\mathbf{r}' \gamma(\mathbf{r}, \mathbf{r}') \hat{h}(\mathbf{r}', \mathbf{r}) + \int d\mathbf{r}_1 d\mathbf{r}_2 \frac{\Gamma(\mathbf{r}_1, \mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|} + E_{\text{nuc}}. \quad (3)$$

The 1- and 2-RDMs carry  $\mathcal{O}(N_g^2)$  and  $\mathcal{O}(N_g^4)$  variables. The exponential wall is gone, and this is the basis of 2-RDM theory. DFT is more ambitious still: it claims that

$$E_0 = E[\rho(\mathbf{r})], \quad (4)$$

a functional of an object with only  $\mathcal{O}(N_g)$  variables. That sounds too good to be true, so we had better prove it.

## 2. Why a Density Could Be Enough (informal DFT)

Split the Hamiltonian into a universal piece and a system-specific piece:

$$\hat{H} = \underbrace{-\frac{1}{2} \sum_i \nabla_i^2 + \sum_{i < j} \frac{1}{r_{ij}}}_{\hat{T} + \hat{V}_{\text{ee}}: \text{ same for every molecule}} + \underbrace{\sum_i v_{\text{ext}}(\mathbf{r}_i)}_{\hat{V}_{\text{ext}}}, \quad v_{\text{ext}}(\mathbf{r}) = - \sum_I \frac{Z_I}{|\mathbf{r} - \mathbf{R}_I|}. \quad (5)$$

Only  $v_{\text{ext}}$  knows what molecule we are studying, and its expectation value is already an explicit density functional,  $E_{\text{ext}} = \int d\mathbf{r} \rho(\mathbf{r}) v_{\text{ext}}(\mathbf{r})$ . So if  $\rho$  determined  $v_{\text{ext}}$ , it would determine  $\hat{H}$ , hence  $\Psi_0$ , hence everything.

The informal argument (due to E. Bright Wilson) is that it obviously does: the integral of  $\rho$  gives  $N$ ; the cusps of  $\rho$  give the nuclear positions  $\mathbf{R}_I$ ; and the slope at each cusp gives the charge,

$$Z_I = - \frac{1}{2\rho(\mathbf{r})} \left. \frac{d\rho(\mathbf{r})}{dr} \right|_{\mathbf{r} \rightarrow \mathbf{R}_I}. \quad (6)$$

So a Coulombic  $v_{\text{ext}}$  is fully specified. Hohenberg and Kohn proved it for *any* external potential.

### 3. The Hohenberg–Kohn Theorems (1964)

**Theorem 1** (HK1). *For a nondegenerate ground state,  $v_{\text{ext}}(\mathbf{r})$  is determined, up to an additive constant, by the ground-state density  $\rho(\mathbf{r})$ .*

*Proof.* Suppose two potentials  $v_1 \neq v_2 + C$  have ground states  $\Psi_1 \neq \Psi_2$  with the *same* density  $\rho$ . The Rayleigh–Ritz principle, applied with  $\Psi_2$  as a trial function for  $\hat{H}_1$ , gives the *strict* inequality (strict because  $\Psi_2$  is not the ground state of  $\hat{H}_1$ )

$$E_1 < \langle \Psi_2 | \hat{H}_1 | \Psi_2 \rangle = \langle \Psi_2 | \hat{H}_2 | \Psi_2 \rangle + \langle \Psi_2 | \hat{H}_1 - \hat{H}_2 | \Psi_2 \rangle = E_2 + \int d\mathbf{r} \rho(\mathbf{r}) [v_1(\mathbf{r}) - v_2(\mathbf{r})]. \quad (7)$$

Exchanging the labels  $1 \leftrightarrow 2$ ,

$$E_2 < E_1 - \int d\mathbf{r} \rho(\mathbf{r}) [v_1(\mathbf{r}) - v_2(\mathbf{r})]. \quad (8)$$

Adding the two lines yields  $E_1 + E_2 < E_1 + E_2$ , a contradiction.  $\square$

**Theorem 2** (HK2). *There exists a functional  $E[\rho]$  that is minimized by the true ground-state density, with the minimum equal to  $E_0$ .*

*Proof.* By HK1 any trial density  $\tilde{\rho}$  comes from a unique  $\tilde{\Psi}$ , so  $E[\tilde{\rho}] = \langle \tilde{\Psi} | \hat{H} | \tilde{\Psi} \rangle \geq E_0 = E[\rho_0]$  by the usual variational principle.  $\square$

Writing the functional as

$$E_{\text{HK}}[\rho] = \underbrace{\int d\mathbf{r} v_{\text{ext}}(\mathbf{r}) \rho(\mathbf{r})}_{\text{system specific}} + \underbrace{F_{\text{HK}}[\rho]}_{\text{universal}}, \quad F_{\text{HK}}[\rho] = \langle \Psi[\rho] | \hat{T} + \hat{V}_{\text{ee}} | \Psi[\rho] \rangle, \quad (9)$$

we would minimize subject to  $\int \rho d\mathbf{r} = N$ ,  $\rho \geq 0$  (part of the  $N$ -representability conditions). One universal functional, computed once, would solve all of chemistry. Unfortunately nobody knows  $F_{\text{HK}}$ , and Levy’s constrained-search construction,

$$E_0 = \min_{\rho \rightarrow N} \left[ \min_{\Psi \rightarrow \rho} \langle \Psi | \hat{T} + \hat{V}_{\text{ee}} | \Psi \rangle + \int d\mathbf{r} v_{\text{ext}} \rho \right], \quad (10)$$

makes it explicit but not computable: the inner search runs over antisymmetric  $N$ -electron wavefunctions.

### 4. The Kohn–Sham Construction (1965)

The practical problem is the kinetic energy. If we write  $E[\rho] = T[\rho] + E_J[\rho] + E'_{\text{xc}}[\rho] + V_{\text{ext}}[\rho]$  with the classical Coulomb (Hartree) term

$$E_J[\rho] = \frac{1}{2} \iint d\mathbf{r} d\mathbf{r}' \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}, \quad (11)$$

then  $T[\rho]$  is both large and unknown. Approximating it directly gives orbital-free DFT, which has never been accurate enough for chemistry.

Kohn and Sham's move: introduce a fictitious system of *noninteracting* electrons with the *same* density. There is always a single determinant reproducing a given well-behaved  $\rho$  (Harriman's construction); for two electrons it is just  $\psi = \sqrt{\rho/2}$ . So write

$$\rho(\mathbf{r}) = \sum_i^{\text{occ}} |\psi_i(\mathbf{r})|^2, \quad T_s = -\frac{1}{2} \sum_i^{\text{occ}} \langle \psi_i | \nabla^2 | \psi_i \rangle, \quad (12)$$

where the  $\psi_i$  are the Kohn–Sham orbitals.  $T_s$  is not  $T$ , but it captures the overwhelming majority of it and is computed exactly from orbitals. Everything we do not know is swept into one term:

$$E_{\text{KS}}[\rho] = T_s[\rho] + E_J[\rho] + V_{\text{ext}}[\rho] + \underbrace{E'_{\text{xc}}[\rho] + (T[\rho] - T_s[\rho])}_{E_{\text{xc}}[\rho]}. \quad (13)$$

Note that  $E_{\text{xc}}$  contains a piece of the *kinetic* energy as well as exchange and correlation. It is the smallest term in the total energy, typically a few percent, which is why crude approximations to it can work. Note also that the KS orbitals are auxiliary objects chosen to reproduce a density; they are not Dyson orbitals and their eigenvalues are not ionization potentials (except for the exact functional's HOMO).

## 5. The Kohn–Sham Equations

Minimizing  $E_{\text{KS}}$  with respect to the orbitals under orthonormality is formally identical to Lecture 1. With  $\psi_i = \sum_{\mu} C_{\mu i} \chi_{\mu}$  and  $\mathcal{L}_{\text{KS}} = E_{\text{KS}} - \sum_{ij} \Lambda_{ij} (\langle i | j \rangle - \delta_{ij})$  we obtain

$$\mathbf{FC} = \mathbf{SC}\boldsymbol{\varepsilon}, \quad \mathbf{F} = \mathbf{h} + \mathbf{J} + \mathbf{V}_{\text{xc}}, \quad v_{\text{xc}}(\mathbf{r}) = \frac{\delta E_{\text{xc}}}{\delta \rho(\mathbf{r})}. \quad (14)$$

The same SCF machinery (orthogonalization, DIIS, Aufbau) applies unchanged. Only the exchange term of Hartree–Fock has been replaced by a multiplicative local potential.

**An explicit functional derivative.** Take the Hartree functional (11). Under  $\rho \rightarrow \rho + \delta\rho$ ,

$$\delta E_J = \frac{1}{2} \iint \frac{\delta \rho(\mathbf{r}) \rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} + \frac{1}{2} \iint \frac{\rho(\mathbf{r}) \delta \rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} = \int d\mathbf{r} \delta \rho(\mathbf{r}) \int d\mathbf{r}' \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}, \quad (15)$$

where the two terms are equal by relabeling. Hence

$$v_{\text{H}}(\mathbf{r}) = \frac{\delta E_J}{\delta \rho(\mathbf{r})} = \int d\mathbf{r}' \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}, \quad (16)$$

the electrostatic potential of the electron cloud. The factor of 1/2 disappears because the functional is quadratic; this is exactly the double counting we saw in the HF orbital energies.

## 6. Where the Functionals Come From: the Uniform Electron Gas (optional)

Consider a box of interacting electrons at uniform density  $\rho_0 = N/V$  with a neutralizing background. This model is solvable, and its energetics can be extracted by dimensional analysis. Under

the coordinate scaling  $\mathbf{r} \rightarrow \lambda \mathbf{r}$  the density must scale as  $\rho_\lambda(\mathbf{r}) = \lambda^3 \rho(\lambda \mathbf{r})$  to conserve  $N$ . Assume a local form  $E[\rho] = C \int d^3r \rho^p$ .

For the kinetic energy, the operator scales as  $\lambda^{-2}$ ; matching powers of  $\lambda$  on both sides of  $\lambda^{-2}E[\rho] = C\lambda^{3-3p} \int \rho^p$  gives  $-2 = 3 - 3p$ , so  $p = 5/3$ : the Thomas–Fermi functional

$$T_s^{\text{TF}}[\rho] = \frac{3}{10}(3\pi^2)^{2/3} \int d^3r \rho^{5/3}(\mathbf{r}). \quad (17)$$

For exchange the operator  $1/r_{12}$  scales as  $\lambda^{-1}$ , giving  $p = 4/3$ : the Dirac–Slater exchange functional

$$E_x^{\text{LDA}}[\rho] = -C_x \int d^3r \rho^{4/3}(\mathbf{r}), \quad C_x = \frac{3}{4} \left( \frac{3}{\pi} \right)^{1/3}. \quad (18)$$

The same result follows from the exact uniform-scaling condition  $E_x[\rho_\lambda] = \lambda E_x[\rho]$ , which any acceptable exchange functional must satisfy. Correlation has no closed form; Ceperley and Alder computed it by diffusion Monte Carlo, and the results were fitted by Vosko–Wilk–Nusair (VWN, 1980) and Perdew–Wang (PW92). Slater exchange plus VWN correlation is the local density approximation, SVWN.

Chemists were brave enough to apply a metal model to molecules, and it worked better than it had any right to: the G2 atomization energy MAE drops from 158.9 kcal/mol (HF) to 90.9 kcal/mol (SVWN). Still useless, but the trend was encouraging enough for Pople to put it in Gaussian.

## 7. Jacob’s Ladder

The general functional we want is nonlocal,  $E[\rho] = \int d^3r f[\rho](\mathbf{r})$ , with  $f$  depending on  $\rho$  everywhere. Perdew’s “Jacob’s ladder” organizes the approximations by what ingredients they use:

- **LDA:**  $f(\rho(\mathbf{r}))$ .
- **GGA:**  $f(\rho, |\nabla\rho|)$ . The canonical example is B88 exchange (Becke, 1988). With the dimensionless reduced gradient  $s(\mathbf{r}) = |\nabla\rho|/\rho^{4/3}$ ,

$$E_x^{\text{B88}} = - \int d^3r \left[ C_x + \frac{\beta s^2}{1 + 6\beta s \sinh^{-1} s} \right] \rho^{4/3}, \quad (19)$$

with a single parameter  $\beta = 0.0042$  fitted to the exchange energies of the rare-gas atoms. It reproduces atomic exchange energies to four digits (e.g. Ne: HF 12.126, Slater 11.043, B88 12.148 a.u.).

- **meta-GGA:** adds  $\nabla^2\rho$  or the kinetic energy density  $\tau(\mathbf{r}) = \sum_i |\nabla\psi_i(\mathbf{r})|^2$ .
- **Hybrids:** mix in exact (HF-like) exchange evaluated with KS orbitals,

$$E_x = (1 - a)E_x^{\text{DFT}}[\rho] + aE_x^{\text{exact}}[\{\psi_i\}]. \quad (20)$$

B3LYP (1993) is the workhorse:

$$E_{\text{xc}}^{\text{B3LYP}} = (1 - a)E_x^{\text{S}} + aE_x^{\text{exact}} + bE_x^{\text{B88}} + cE_c^{\text{LYP}} + (1 - c)E_c^{\text{VWN}}. \quad (21)$$

Why not 100% exact exchange? Because DFT correlation functionals are built to cancel errors in DFT exchange; the adiabatic connection,  $E_{\text{xc}} = \int_0^1 d\lambda \langle V_{\text{ee}} \rangle_\lambda$ , gives the formal justification for a fraction near 1/4.

- **Double hybrids:** add MP2-like correlation (Lecture 3).

Note the ladder is not monotone: climbing costs more and does not guarantee improvement, since every rung contains fitted forms.

## 8. Numerical Integration (optional)

None of these integrals can be done analytically, so  $E_{\text{xc}}$  is evaluated on an atom-centered quadrature grid (Becke partitioning of space, radial  $\times$  Lebedev angular grids):

$$E_{\text{x}}^{\text{grid}} = -C_{\text{x}} \sum_k w_k \rho_k^{4/3}. \quad (22)$$

This is  $\mathcal{O}(N)$  work and is not the bottleneck; the  $\mathcal{O}(N^3)$  diagonalization is. But grids introduce a real error. Perturb the grid densities by  $\delta\rho_k$ :

$$\delta E_{\text{x}}^{\text{grid}} = -\frac{4}{3} C_{\text{x}} \sum_k w_k \rho_k^{1/3} \delta\rho_k + \mathcal{O}(\delta\rho^2). \quad (23)$$

The error is weighted by  $\rho_k^{1/3}$ , whereas conserving the electron number constrains only the unweighted sum,  $\sum_k w_k \delta\rho_k = 0$ . A grid that integrates the number of electrons exactly can therefore still give a poor exchange energy: errors in high-density (core) regions are weighted more heavily than the number constraint sees. This is why grid quality must be converged separately, and why energy differences on coarse grids can show spurious structure.

## 9. Exact Conditions and the Self-Interaction Error

Since we cannot derive the exact functional, we constrain it. The most instructive constraint concerns a single electron. For a normalized one-electron density there is no electron–electron interaction at all, so the Hartree term must be cancelled exactly by exchange, and correlation must vanish:

$$E_{\text{H}}[\rho] + E_{\text{x}}[\rho] + E_{\text{c}}[\rho] = 0 \quad (N = 1). \quad (24)$$

In Hartree–Fock this holds identically, because  $K_{ii} = J_{ii}$  for one occupied orbital. Approximate functionals fail it:

| Method | $E_{\text{J}} + E_{\text{xc}}$ for the H atom (a.u.) |
|--------|------------------------------------------------------|
| HF     | 0                                                    |
| SVWN   | 0.00277                                              |
| BLYP   | 0.00140                                              |
| B3LYP  | −0.00281                                             |

This *self-interaction error* (equivalently, delocalization error) is the single largest source of qualitative DFT failures:

- $\text{H}_2^+$  should dissociate to  $\text{H} + \text{H}^+$ . SVWN instead predicts two half-charged fragments, in error by 49 kcal/mol. HF gets this right.

- Reaction barriers are systematically underestimated; the transition state has a delocalized, stretched-bond density that SIE stabilizes. For  $\text{H}_2 + \text{H} \rightarrow \text{H} + \text{H}_2$  the error is  $\sim 9$  kcal/mol.
- Charge-transfer energetics and band gaps are badly predicted.

The standard partial fix is a range-separated hybrid. Split the interaction with the error function,

$$\frac{1}{r} = \underbrace{\frac{\text{erfc}(\omega r)}{r}}_{\text{short range}} + \underbrace{\frac{\text{erf}(\omega r)}{r}}_{\text{long range}}, \quad (25)$$

and use DFT exchange at short range, exact exchange at long range. This removes the long-range SIE by construction; in  $\omega\text{B97X-V}$ , for instance, the H-atom error drops to  $-0.00079$  a.u. and the  $\text{H}_2^+$  dissociation error to 20 kcal/mol. Better, not solved.

## 10. Two More Failures

**Strong correlation.** KS-DFT builds the density from a single determinant. When several determinants are equally important (stretched  $\text{H}_2$ , where the exact wavefunction becomes  $c_1|\sigma_g^2\rangle + c_2|\sigma_u^2\rangle$  with  $c_1 \rightarrow -c_2$ ; antiferromagnetically coupled metal centers; mixed-valence compounds), no approximate  $E_{\text{xc}}[\rho]$  in current use repairs this. The exact functional would, but we have no constructive route to it. In practice one breaks spin symmetry and lives with the consequences.

**Dispersion.** London dispersion is a correlation effect between instantaneous fluctuating dipoles on *distant* fragments, decaying as  $-C_6/R^6$ . A semilocal functional evaluated at  $\mathbf{r}$  has no way of knowing about density fluctuations far away, and exact exchange does not help because dispersion is a correlation effect. The pragmatic fix (Grimme, 2004) is an additive atom-pairwise correction,

$$E_{\text{DFT-D}} = E_{\text{DFT}} - \sum_{A < B} f_{\text{damp}}(R_{AB}) \frac{C_6^{AB}}{R_{AB}^6}, \quad (26)$$

with damping at short range to avoid double counting. Nonlocal correlation kernels (VV10) do the same job from the density. This problem is essentially solved in practice.

## 11. Cost, Accuracy, and What to Take Away

DFT costs what HF costs:  $\mathcal{O}(N^3)$  asymptotically,  $\sim 5000$  basis functions and 500 electrons overnight on a node. Accuracy on general thermochemistry benchmarks (weighted MAD, GMTKN-type sets, in kcal/mol):

| Method | WTMAD (kcal/mol) |
|--------|------------------|
| HF     | 18.5             |
| SVWN   | 6.9              |
| BLYP   | 5.3              |
| M06-L  | 4.8              |
| B3LYP  | 3.7              |

So DFT buys a factor of five over HF for free. Three things to remember:

1. DFT is exact in principle and approximate in practice; all the physics is hidden in  $E_{xc}$ , which is small in magnitude but decisive.
2. Its systematic failures, SIE, strong correlation and dispersion, are understood well enough that you can predict when not to trust it.
3. It is not systematically improvable. If you need a better answer, you must change frameworks, and go back to the wavefunction.

That is Lecture 3.